

Neural-Inspired Multi-Agent Molecular Communication Networks for Collective Intelligence

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1. Motivation & Introduction

The **Internet of Bio-Nano Things (IoBNT)** envisions nanomachine networks embedded inside biological environments for real-time monitoring and actuation. Molecular Communication (MC) is the canonical channel paradigm, yet current theoretical models rest on two problematic assumptions:

- **SISO / structured topology:** Most works study point-to-point or simple geometric MIMO links — unrepresentative of randomly distributed nanomachines in living tissue.
- **Super-capable individual agents:** Models assume perfect synchronisation, high-precision molecule counting, and complex on-board signal processing — far beyond current nanotechnology.

Biological insight: The cortex does not compute because neurons are individually intelligent. It computes because *millions of simple threshold-firing units act collectively.*

We propose importing this principle into MC:

“Super-capable individual agents”



“Network of simple, numerous agents”

2. System Model

Setup: M spherical transceiver agents (radius a_{rx}) distributed by a Poisson point process inside a 3-D volume. Molecular motion follows *free diffusion* with isotropic, homogeneous coefficient D . Time is slotted into intervals of duration T_s . *Fully absorbing* receivers are adopted for physical plausibility (surface absorption leaves interior volume for storage/computation).

Cumulative absorption probability — the probability that a molecule emitted by agent j is absorbed by agent i by time t :

$$g_{ij}(t) = \frac{a_{rx}}{d_{ij}} \operatorname{erfc}\left(\frac{d_{ij} - a_{rx}}{\sqrt{4Dt}}\right), \quad (1)$$

where $d_{ij} = |\mathbf{r}_i - \mathbf{r}_j|$. The per-interval absorption probability is $P_{ij}[\ell] = g_{ij}(\ell T_s) - g_{ij}((\ell-1)T_s)$.

Gaussian arrival model. Molecule counts are approximated as Gaussian (standard in MC for large N_0). Conditioned on the binary state vector $\mathbf{x}[k] \in \{0, 1\}^M$ over memory horizon L :

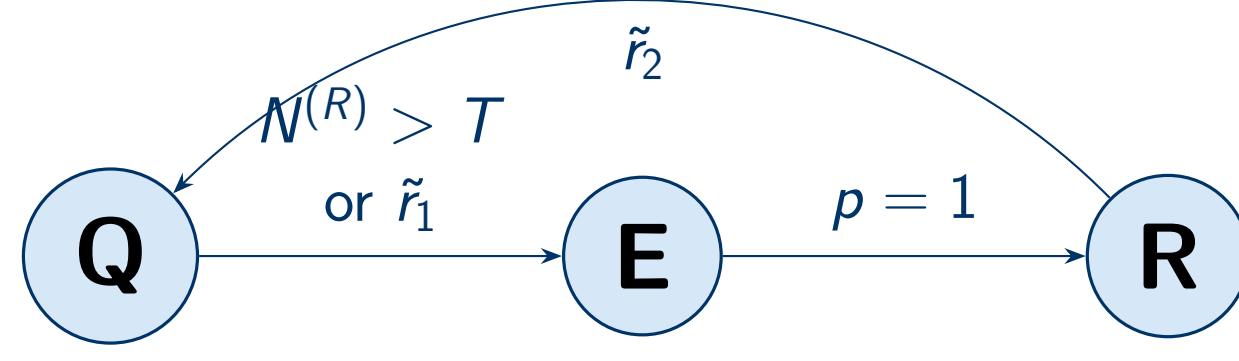
$$\mu_i[k] = N_0 \sum_{\ell=1}^L \sum_j P_{ij}[\ell] x_j[k - \ell], \quad (2)$$

$$\sigma_i^2[k] = N_0 \sum_{\ell,j} P_{ij}[\ell] (1 - P_{ij}[\ell]) x_j[k - \ell]. \quad (3)$$

Fig. 1: Proposed swarm of excitable nanomachines communicating via diffusion (contrasted with traditional SISO MC).

3. Greenberg–Hastings Excitable Agent Model

We adopt the **Greenberg–Hastings (GH) cellular automaton** to model agent dynamics. Each agent occupies one of three discrete states:



Quiescent state (Q): Resting state; fires if absorbed molecules exceed threshold T , or spontaneously w.p. $\tilde{r}_1$.

Excited state (E): Releases N_0 molecules isotropically; transitions to R with probability 1.

Refractory state (R): Recovery; returns to Q w.p. $\tilde{r}_2$.

Excitation probability at step k for agent i :

$$\gamma_i[k] = \Pr\left\{N_i^{(R)}[k] > T\right\} = 1 - \Phi\left(\frac{T - \mu_i[k]}{\sigma_i[k]}\right), \quad (4)$$

where $\Phi(\cdot)$ is the standard normal CDF.

The state-evolution equations (derived from the GH transition rules) are:

$$r_i[k+1] = e_i[k] + (1 - \tilde{r}_2) r_i[k], \quad (5)$$

$$e_i[k+1] = (\tilde{r}_1 + \gamma_i[k]) q_i[k]. \quad (6)$$

4. Mean-Field Steady-State Analysis

As $k \rightarrow \infty$ the state probabilities converge: $\{e_i, r_i, q_i\}[k] \rightarrow \{e^*, r^*, q^*\}$. Setting $\mathbf{e}[k] = \mathbf{e}[k+1] = \mathbf{e}^*$ and using $e_j[k] \approx e_j^*$ (mean-field independence), the **unconditional mean and variance** of received molecules become:

$$\boldsymbol{\mu}^* = N_0 \left(\sum_{\ell=1}^L \mathbf{P}[\ell] \right) \mathbf{e}^*, \quad (7)$$

$$\boldsymbol{\sigma}^{*2} = N_0 \sum_{\ell=1}^L \left(\mathbf{P}[\ell] \odot (1 - \mathbf{P}[\ell]) \right) \mathbf{e}^* + N_0^2 \sum_{\ell=1}^L \mathbf{P}^2[\ell] \mathbf{e}^* \odot (1 - \mathbf{e}^*), \quad (8)$$

where $\odot$ is element-wise multiplication.

These expressions reduce the M -agent network to the following **M -dimensional fixed-point problem**:

$$\mathbf{e}^* = \frac{\tilde{r}_1 + \boldsymbol{\Gamma}(\mathbf{e}^*)}{1 + (\tilde{r}_1 + \boldsymbol{\Gamma}(\mathbf{e}^*)) \left(1 + \frac{1}{\tilde{r}_2}\right)}, \quad \boldsymbol{\Gamma}(\mathbf{e}^*) = 1 - \Phi\left(\frac{T - \boldsymbol{\mu}^*}{\boldsymbol{\sigma}^*}\right). \quad (9)$$

Solved via **under-relaxed fixed-point iteration**:

$$\mathbf{e}_{k+1}^* = \lambda \mathbf{g}(\mathbf{e}_k^*) + (1 - \lambda) \mathbf{e}_k^*, \quad \lambda = 0.9, \varepsilon = 10^{-12}.$$

Physical interpretation of divergence near T_c : The mean-field assumption requires statistical independence between nodes. At the critical point, long-range correlations develop and the independence assumption breaks down — explaining the observable theory-simulation gap exactly at the transition.

5. Results: Phase Transition & Critical Regime

Simulation parameters:

Parameter	Value	Parameter	Value
Agents (N)	100	Receiver radius (a_{rx})	$4 \mu\text{m}$
Medium radius	$20 \mu\text{m}$	Diff. coeff. (D)	$79.4 \mu\text{m}^2/\text{s}$
Emission (N_0)	100 molecules	Memory (L)	5
Time step (T_s)	1 s	$\tilde{r}_2$	0.5

As Q_{th} increases, mean activity drops sharply near $Q_{\text{th}} \approx 550$. The **standard deviation peaks at the transition**, quantifying maximum network susceptibility:

Fig. 2: Mean and std. dev. of excited activity vs. activation threshold Q_{th} .

Regime	Threshold	Behaviour
Sub-critical (saturation)	$Q_{\text{th}} < 450$	Noise dominates
Critical (optimal)	$Q_{\text{th}} \approx 500\text{--}550$	Max. info. transfer
Super-critical (silence)	$Q_{\text{th}} > 600$	Activity dies out

6. Results: Mutual Information & Model Validation

MI metrics quantify statistical coupling across the swarm:

$$\bar{I}_{\text{pair}} = \binom{M}{2}^{-1} \sum_{i < j} I(X_i; X_j), \quad \bar{I}_{\text{coll}} = \frac{1}{N} \sum_{i=1}^N I(X_i; M_i), \quad (10)$$

where M_i is the aggregate activity of all agents excluding i . Both metrics **peak precisely at T_c** :

Fig. 3: Pairwise and collective MI both peak at the critical threshold $T_c=550$, confirming collective information processing is maximised there.

The fixed-point prediction closely matches stochastic simulation across all threshold values; divergence appears *only* at T_c , exactly where inter-agent correlations are strongest:

Fig. 4: Theoretical (mean-field) vs. simulated excitation probability. Inset: absolute error (note log scale).

7. Conclusions & Future Work

- A **decentralised swarm** of simple nanomachines can achieve rich collective dynamics via diffusion alone — no complex individual processing required.
- Tuning a single parameter — the **activation threshold** — drives the network to a critical regime where information propagation is maximised.
- A **closed-form mean-field fixed-point framework** accurately predicts macroscopic behaviour outside the transition region.
- Pairwise and collective **MI metrics peak at criticality**, providing a quantitative signature of collective intelligence.

Future Directions:

- Systematic **parameter sensitivity analysis** (density, diffusion coefficient, network geometry).
- Define external input-output channel model for rigorous **information-theoretic capacity analysis**.
- Extend analytical framework to **account for inter-node correlations** near T_c (finite-size scaling).
- Relax isotropic diffusion assumption to model **heterogeneous in-body tissue** environments.

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